

Transformer Fine-Tuning for Political Question Evasion Detection in CLARITY

Project-MMD2 Team

Abstract

Political interviews often contain answers that are topically related to a question while avoiding the requested commitment. We describe a transformer fine-tuning system for the SemEval 2026 CLARITY shared task on political question evasion detection. The system addresses two settings: clarity classification into *Clear Reply*, *Ambivalent*, and *Clear Non-Reply*, and fine-grained strategy classification over nine reply and evasion strategies. We train single-task classifiers with BERT-base, RoBERTa-large, DeBERTa-v3-base, DeBERTa-v3-large, and DeBERTa-xlarge using a simple question-answer input format, and compare class-weighted training, an input wording ablation, and a shared-encoder multi-task model. On the saved runs, DeBERTa-xlarge gives the strongest Task 1 test result with 0.632 macro-F1, while RoBERTa-large gives the strongest Task 2 majority-consensus test result with 0.342 macro-F1. Unanimous Task 2 examples are consistently easier than majority-consensus examples, indicating that annotator disagreement is a central difficulty. Overall, strong encoders provide useful baselines, but fine-grained political evasion strategy detection remains challenging under class imbalance and pragmatic ambiguity.

1 Introduction

Political question answering is not only a problem of topical relevance. A response can mention the same actors, policies, or events as the question while declining to answer the requested point. Speakers may answer only part of a question, shift to a broader talking point, reject a premise, cite uncertainty, or provide a direct answer inside a long digression. Detecting these phenomena matters for computational political communication because semantic similarity alone does not distinguish a clear answer from a strategic non-answer.

The CLARITY shared task studies this problem with political question-answer pairs (cla, 2026; qev, 2026). It

follows prior work on response clarity classification and evasion taxonomies, including “I Never Said That” (ine, 2024). The task has two levels. Task 1 asks whether an answer is a *Clear Reply*, *Ambivalent*, or *Clear Non-Reply*. Task 2 asks for a fine-grained strategy label, such as *Explicit*, *Implicit*, *Dodging*, *Deflection*, or *Declining to answer*.

We build a reproducible Hugging Face Transformers system (Wolf et al., 2020) for both tasks. Our goal is to establish clean baselines and error analyses rather than to introduce a complex architecture. We compare several encoder families, evaluate simple task-specific changes, and analyze how model behavior changes under label imbalance and annotator disagreement. The main contributions are:

1. a complete single-task training and evaluation pipeline for both CLARITY tasks;
2. a comparison of BERT, RoBERTa, DeBERTa-v3, and DeBERTa-xlarge encoders (Devlin et al., 2019; Liu et al., 2019; He et al., 2021b,a);
3. controlled comparisons of class weighting, input wording, model scale, and shared-encoder multi-task learning;
4. per-class, confusion, confidence, length, and annotator-disagreement analyses.

2 Task and Dataset

We use the Hugging Face dataset `ailsntua/QEvasion`. The repository expects the columns `question`, `interview_answer`, `clarity_label`, `evasion_label`, `annotator1`, `annotator2`, `annotator3`, and `index`. The official training split provides labels for both tasks. The official test split provides one clarity label for Task 1 and three annotator labels for Task 2.

All runs use an 80/20 internal split of the official training data, giving 2,758 training examples and 690

Task	Split	N	Majority label		Smallest label	
Task 1	Dev	690	Ambivalent (59.1%)	408	Clear Non-Reply 71 (10.3%)	
Task 1	Test	308	Ambivalent (66.9%)	206	Clear Non-Reply 23 (7.5%)	
Task 2	Dev	690	Explicit (30.6%)	211	Partial/half-answer 16 (2.3%)	
Task 2	Test maj.	275	Explicit (28.0%)	77	Partial/half-answer 3 (1.1%)	

Table 1: Dataset split and label-imbalance summary from the saved split and prediction files, consistent with the distribution tables in `paper_assets/tables/`. The Task 2 test row uses majority-consensus labels.

development examples. The test split has 308 examples. The split is stratified where possible: single-task Task 1 runs stratify by clarity label, single-task Task 2 runs stratify by evasion label, and the multi-task run stratifies by clarity label. All saved experiments use seed 42.

Table 1 summarizes the label imbalance. Task 1 is dominated by *Ambivalent*. Task 2 is more imbalanced: the largest classes are *Explicit*, *Dodging*, and *Implicit*, while *Partial/half-answer*, *Clarification*, and *Claims ignorance* are rare. For Task 2 test evaluation, 275 of 308 examples have a majority label, 125 are unanimous, and 33 have no majority.

3 System Description

Input format. The main input template concatenates the normalized question and interview answer:

```
Question: {question} Answer:
{answer}
```

An input wording ablation replaces `Answer:` with `Politician's answer:.` We do not use a natural-language-inference template.

Single-task classifiers. For each task, we fine-tune `AutoModelForSequenceClassification`. Task 1 uses a three-way classifier and Task 2 uses a nine-way classifier. The implemented pipeline saves the config snapshot, split indices and IDs, label mapping, tokenizer, best model, training history, dev and test metrics, predictions, probabilities, logits, classification reports, confusion matrices, and misclassified examples.

Class-weighted loss. The baseline loss is cross-entropy. Class-weighted experiments use inverse-frequency weights computed from the internal training

split. For Task 1, the weights are 1.093 for *Clear Reply*, 0.563 for *Ambivalent*, and 3.226 for *Clear Non-Reply*. For Task 2, the largest weights are assigned to the rare classes *Partial/half-answer*, *Clarification*, *Claims ignorance*, and *Declining to answer*.

Multi-task classifier. The multi-task model uses a shared transformer encoder with two linear heads. One head predicts the Task 1 clarity label and the other predicts the Task 2 strategy label. The training loss is

$$L = L_{\text{task1}} + \lambda_{\text{task2}} L_{\text{task2}},$$

with $\lambda_{\text{task2}} = 1.0$ in the saved configuration. The best checkpoint is selected by Task 2 development macro-F1.

Configurations. Table 2 summarizes the YAML configurations used in the saved experiments. Shared training defaults include weight decay 0.01, warmup ratio 0.1, maximum gradient norm 1.0, epoch-level evaluation and checkpointing, best-checkpoint loading by validation metric, early stopping patience 2, and one saved checkpoint. DeBERTa-v3 tokenizers are loaded with `use_fast=false` in the configs.

4 Experimental Setup

Training uses `Hugging Face Trainer`. Models are evaluated on the internal development split after each epoch, and the best checkpoint is loaded at the end. The primary metric is macro-F1. We also report weighted-F1 and accuracy. The analysis assets compute 95% bootstrap confidence intervals over examples for the main macro-F1 values; these intervals do not measure training-seed variance.

Task 1 test evaluation uses the single clarity label. Task 2 test evaluation uses three annotator labels. We report majority-consensus macro-F1 on examples with at least two agreeing annotators, unanimous-only macro-F1 on examples where all three annotators agree, and any-annotator match rate over all 308 test examples. Rows without a majority are treated as unresolved annotation disagreement rather than as standard majority-label errors.

5 Results

5.1 Task 1: Clarity Classification

Table 3 reports Task 1 results. DeBERTa-v3-large has the highest development macro-F1 at 0.690, while

Config family	Encoder	Max len.	Epochs	Train/Eval bs	Accum.	LR	Precision and memory settings
BERT-base T1/T2	bert-base-uncased	256	5	16/32	1	2e-5	fp32
DeBERTa-v3-base T1/T2	microsoft/deberta-v3-base	256	6	8/16	2	1.5e-5	fp16
DeBERTa-v3-base weighted	microsoft/deberta-v3-base	256	6	8/16	2	1.5e-5	fp16, class weights
DeBERTa-v3-base alt input	microsoft/deberta-v3-base	256	6	8/16	2	1.5e-5	fp16, alternate prefix
RoBERTa-large T1/T2	roberta-large	256	5	2/4	8	1e-5	fp16, grad. checkpointing
DeBERTa-v3-large T1/T2	microsoft/deberta-v3-large	256	5	2/4	8	1e-5	fp16
DeBERTa-xlarge T1/T2	microsoft/deberta-xlarge	192	4	1/2	16	8e-6	fp32, grad. checkpointing
DeBERTa-v3-base multi-task	microsoft/deberta-v3-base	256	6	8/16	2	1.5e-5	fp16, $\lambda_{\text{task2}} = 1.0$

Table 2: Configuration summary for the saved runs. “Accum.” is gradient accumulation steps; train/eval batch sizes are per device.

DeBERTa-xlarge has the highest test macro-F1 at 0.632 with a 95% bootstrap interval of 0.554–0.692. Figure 1 shows the same test comparison. The ranking suggests that the largest model helps on the held-out test set, but the differences among RoBERTa-large, DeBERTa-v3-large, and DeBERTa-v3-base are modest.

5.2 Task 2: Fine-Grained Strategy Classification

Table 4 reports Task 2 results. RoBERTa-large is best on both development and majority-consensus test macro-F1, with 0.375 on development and 0.342 on test. Its test bootstrap interval is 0.239–0.402. DeBERTa-v3-large is second by majority macro-F1 and has the highest majority accuracy. Figure 2 shows that all systems remain below 0.35 majority-consensus macro-F1 on the test set, indicating that the fine-grained strategy task is substantially harder than the coarse clarity task.

Annotator agreement changes the interpretation of Task 2 results. Figure 3 compares majority-consensus and unanimous-only evaluation. Unanimous-only macro-F1 is higher for every saved system. RoBERTa-large obtains 0.342 majority macro-F1 and 0.402 unanimous macro-F1; class-weighted DeBERTa-v3-base obtains the highest unanimous-only score, 0.426, but has lower majority accuracy and any-annotator match rate. DeBERTa-v3-large has the highest any-annotator match rate at 0.542.

6 Ablation and Analysis

Class weighting. Class weighting has different effects across tasks. For Task 1, weighted DeBERTa-v3-base lowers test macro-F1 from 0.605 to 0.546. For Task 2, it raises DeBERTa-v3-base majority-consensus test macro-F1 from 0.301 to 0.318, but lowers majority accuracy from 0.400 to 0.280. This pattern suggests that class weighting increases attention to minority strategies but

can also increase false positives.

Input wording. The alternate prefix *Politician’s answer:* does not improve the saved runs. For Task 1, DeBERTa-v3-base drops from 0.605 to 0.588 test macro-F1. For Task 2, it drops from 0.301 to 0.238 majority-consensus test macro-F1. The shorter *Answer:* prefix is therefore retained.

Model scale. Increasing model size helps Task 1 more than Task 2. DeBERTa-xlarge is best for Task 1 test macro-F1, but RoBERTa-large is best for Task 2. DeBERTa-v3-large is competitive on Task 2 and has the highest any-annotator match rate, but the xlarge DeBERTa run does not improve the fine-grained strategy metric.

Multi-task learning. The shared-encoder model improves Task 1 development macro-F1 over single-task DeBERTa-v3-base, 0.676 versus 0.664, but lowers Task 1 test macro-F1 to 0.586. For Task 2, it is below the single-task DeBERTa-v3-base run on both development and majority-consensus test macro-F1. The simple equal-weight objective is therefore not sufficient to make the two tasks help each other reliably.

7 Error Analysis

The best Task 1 system, DeBERTa-xlarge, performs best on the majority class. Its per-class test F1 is 0.765 for *Ambivalent*, 0.576 for *Clear Reply*, and 0.553 for *Clear Non-Reply*. The most common errors are *Ambivalent* predicted as *Clear Reply* (47 examples), *Clear Reply* predicted as *Ambivalent* (26 examples), and *Clear Non-Reply* predicted as *Ambivalent* (10 examples). Figure 4 shows that non-replies are often softened into the ambiguous middle category.

System	Dev macro-F1	Test macro-F1	Test weighted-F1	Test acc.
BERT-base	0.576	0.570	0.671	0.692
DeBERTa-v3-base	0.664	0.605	0.688	0.688
DeBERTa-v3-base + weighted	0.631	0.546	0.656	0.653
DeBERTa-v3-base + alt input	0.671	0.588	0.684	0.679
DeBERTa-v3-base multi-task	0.676	0.586	0.694	0.688
RoBERTa-large	0.677	0.596	0.671	0.669
DeBERTa-v3-large	0.690	0.595	0.699	0.698
DeBERTa-xlarge	0.687	0.632	0.701	0.695

Table 3: Task 1 development and test results. The main metric is macro-F1.

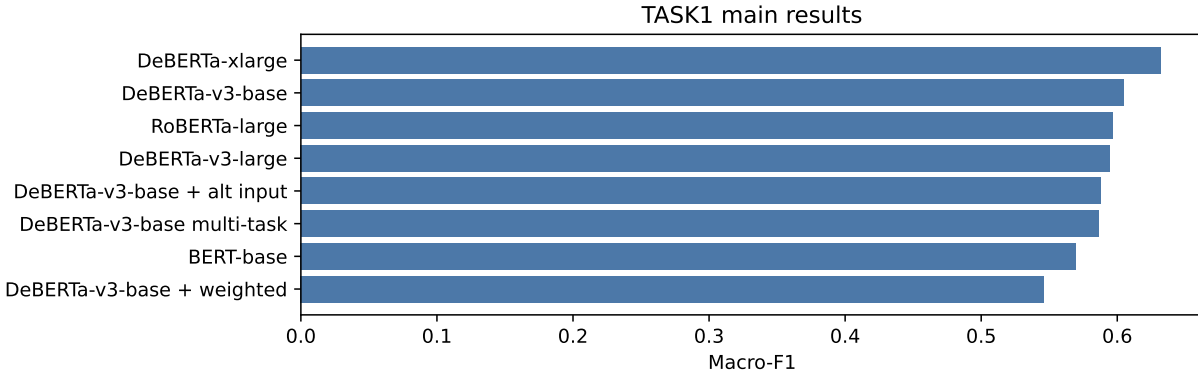


Figure 1: Task 1 test macro-F1 across saved systems. DeBERTa-xlarge is strongest on the test set, while class-weighted DeBERTa-v3-base is weaker than the unweighted DeBERTa-v3-base baseline.

For Task 2, RoBERTa-large has very uneven per-class behavior. Figure 5 shows high F1 for *Clarification* and *Explicit*, but 0.000 F1 for *Partial/half-answer*, 0.127 for *Implicit*, and 0.133 for *General*. The low-support classes should be interpreted cautiously, but the pattern aligns with the confusion table: *Implicit* is often predicted as *Explicit* or *Dodging*, and *General* is often predicted as *Dodging* or *Explicit*. Figure 6 visualizes these confusions.

Table 5 gives representative qualitative error patterns from the saved example files. The examples show that errors often arise when an answer contains both a direct local response and broader strategic framing.

8 Additional Analysis

Length and truncation. The saved length analysis uses approximate whitespace word counts. Test examples average 344 total words, with about 23.4% marked as likely truncated by the analysis heuristic. In Task 2, difficult categories such as *General*, *Implicit*, and *Partial/half-answer* have long average answer lengths in the truncation table. This supports the error analy-

sis: long answers often mix a partial answer with background, justification, or topic shifts.

Confidence. Confidence separates correct and incorrect predictions only partially. For the best Task 1 test system, DeBERTa-xlarge, mean maximum probability is 0.837 overall, 0.861 on correct predictions, and 0.784 on wrong predictions. For the best Task 2 test system, RoBERTa-large, the corresponding values are 0.401, 0.460, and 0.364. These gaps are useful, but high-confidence wrong examples remain.

Annotator disagreement. Task 2 disagreement is substantial: 33 of 308 test rows have no majority label. Majority-consensus evaluation gives a single reference label for 275 examples, while unanimous-only evaluation covers 125 high-agreement examples. The consistent increase from majority to unanimous macro-F1 in Figure 3 suggests that model difficulty is partly tied to human label uncertainty.

System	Dev macro-F1	Test maj. macro-F1	Test weighted-F1	Test maj. acc.
BERT-base	0.335	0.239	0.255	0.305
DeBERTa-v3-base	0.300	0.301	0.334	0.400
DeBERTa-v3-base + weighted	0.331	0.318	0.278	0.280
DeBERTa-v3-base + alt input	0.277	0.238	0.308	0.378
DeBERTa-v3-base multi-task	0.280	0.262	0.354	0.404
RoBERTa-large	0.375	0.342	0.338	0.385
DeBERTa-v3-large	0.368	0.325	0.380	0.418
DeBERTa-xlarge	0.336	0.267	0.356	0.400

Table 4: Task 2 development and test results. Test scores use majority-consensus labels on the 275 test examples where at least two annotators agree.

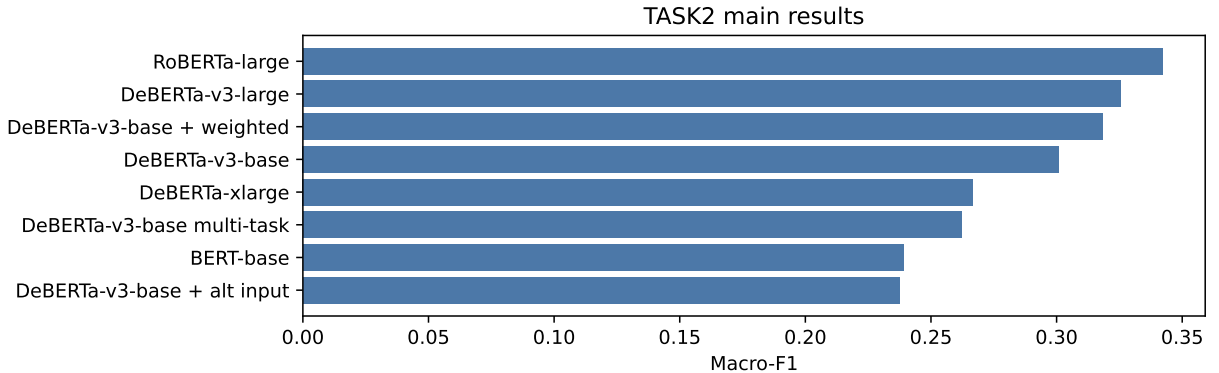


Figure 2: Task 2 majority-consensus test macro-F1 across saved systems. RoBERTa-large is the strongest saved model, followed by DeBERTa-v3-large and class-weighted DeBERTa-v3-base.

9 Limitations

The saved experiments use a single seed per configuration, so model comparisons do not estimate training-seed variance. The bootstrap intervals quantify example-level uncertainty only. The dataset is imbalanced, especially for minority Task 2 strategies, and many political answers are pragmatically ambiguous even for human annotators. The large-model configurations are constrained by GPU memory, using small per-device batches, gradient accumulation, and in some cases shorter maximum sequence length. The length and truncation analyses use whitespace-based approximations rather than tokenizer-specific token counts.

10 Conclusion

We present a reproducible transformer fine-tuning system for the CLARITY shared task on political question evasion detection. DeBERTa-xlarge is the strongest saved Task 1 model with 0.632 test macro-F1, while RoBERTa-large is the strongest saved Task 2 model

with 0.342 majority-consensus test macro-F1. Class weighting helps the fine-grained Task 2 macro metric for DeBERTa-v3-base but hurts Task 1, and the simple shared-encoder multi-task model does not consistently outperform single-task training. Future work should evaluate multiple seeds, stronger hierarchical models, calibrated decision rules, rationale-aware supervision, prompt-based large language model systems, and training objectives that model annotator disagreement directly.

References

- “i never said that”: A dataset, taxonomy and baselines on response clarity classification, 2024. Bibliographic metadata pending.
- CLARITY: Semeval 2026 shared task on unmasking political question evasions. <https://konstantinosftw.github.io/CLARITY-SemEval-2026/>, 2026. Shared-task website; official task-paper metadata pending.

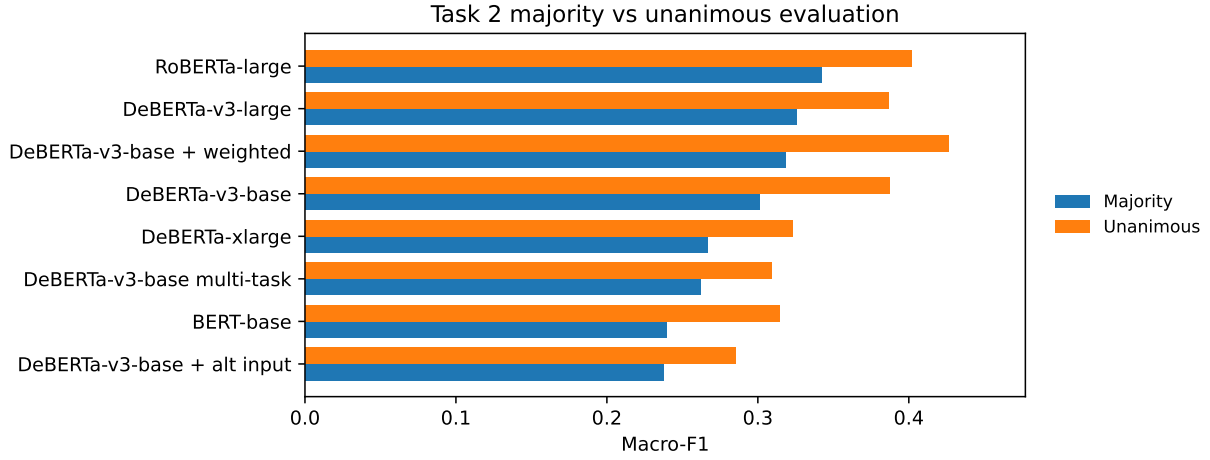


Figure 3: Task 2 majority-consensus versus unanimous-only macro-F1. Higher unanimous-only scores show that high-agreement examples are easier for the classifiers.

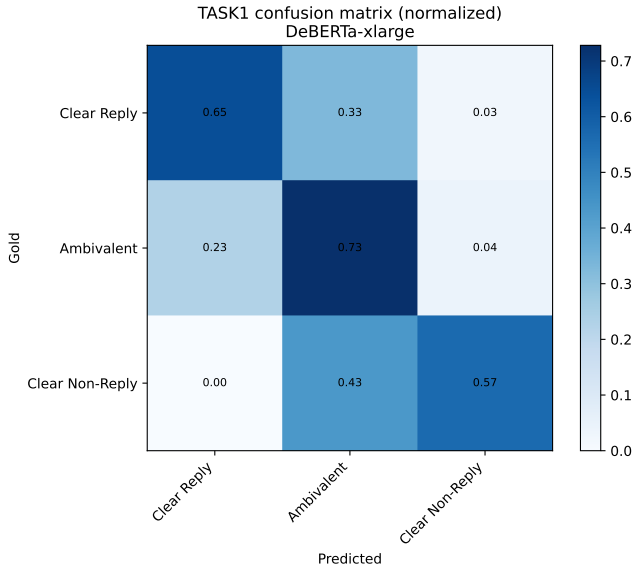


Figure 4: Normalized Task 1 confusion matrix for DeBERTa-xlarge on test. The largest off-diagonal errors involve *Ambivalent* versus *Clear Reply* and *Clear Non-Reply* versus *Ambivalent*.

QEvasion: Political question evasion dataset. <https://huggingface.co/datasets/ailsntua/QEvasion>, 2026. Hugging Face dataset card; citation metadata pending.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*,

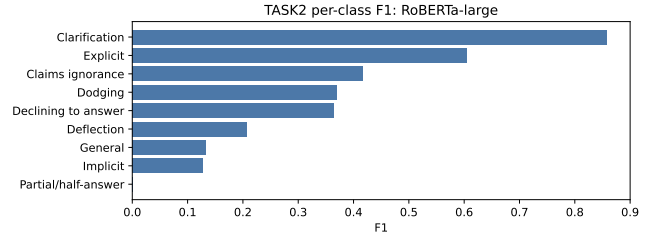


Figure 5: Per-class Task 2 F1 for RoBERTa-large on majority-consensus test examples. Fine-grained labels differ sharply in difficulty.

pages 4171–4186, 2019.

Pengcheng He, Jianfeng Gao, and Weizhu Chen. DeBERTaV3: Improving DeBERTa using ELECTRA-style pre-training with gradient-disentangled embedding sharing. *arXiv preprint arXiv:2111.09543*, 2021a.

Pengcheng He, Xiaodong Liu, Jianfeng Gao, and Weizhu Chen. DeBERTa: Decoding-enhanced BERT with disentangled attention. In *International Conference on Learning Representations*, 2021b.

Yinhan Liu, Myle Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, and Veselin Stoyanov. RoBERTa: A robustly optimized BERT pretraining approach. *arXiv preprint arXiv:1907.11692*, 2019.

Thomas Wolf, Lysandre Debut, Victor Sanh, Julien Chaumond, Clement Delangue, Anthony Moi, Pierric Cistac, Tim Rault, Remi Louf, Morgan Funtowicz,

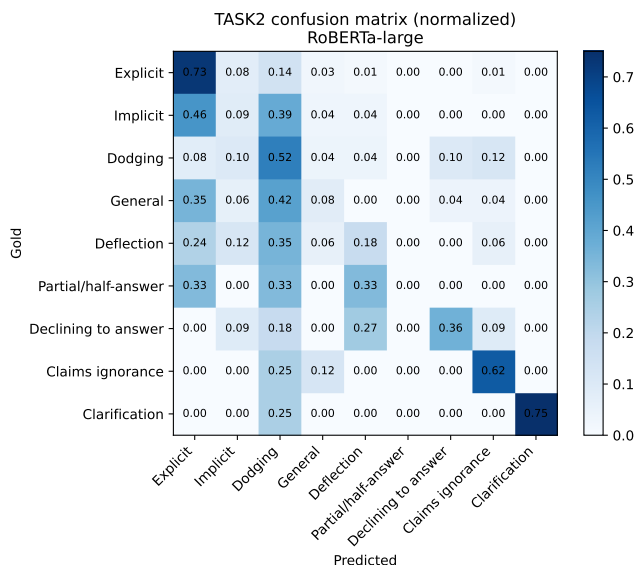


Figure 6: Normalized Task 2 confusion matrix for RoBERTa-large on majority-consensus test examples. Many errors concentrate around *Explicit*, *Dodging*, and *Implicit*.

Joe Davison, Sam Shleifer, Patrick von Platen, Clara Ma, Yacine Jernite, Julien Plu, Canwen Xu, Teven Le Scao, Sylvain Gugger, Mariama Drame, Quentin Lhoest, and Alexander M. Rush. Transformers: State-of-the-art natural language processing. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 38–45, 2020.

Task	Gold → prediction	Pattern
Task 1	Ambivalent Clear Reply	→ A direct rejection is followed by extended framing about violence and governance.
Task 1	Clear Non-Reply → Ambivalent	The answer refuses a hypothetical question but still gives related policy context.
Task 2	Implicit → Explicit	The answer gives a plausible policy explanation without isolating the requested reason.
Task 2	General → Dodging	The response shifts across process, regional actors, and another speaker.

Table 5: Representative error patterns from `paper_assets/examples/`. The table summarizes examples rather than reproducing long answers.